Height Growth Equations for CONUS

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Data

We took the FIA height growth data, filtered out FIA plot locations where the diameter at the end of the 5-year or greater measurement interval was less than the starting diameter. This left 3495112 growth observations.

The observations are distributed among species codes as follows (pre:

Species	Common Name	n
131	loblolly pine	401537
316	red maple	222079
202	Douglas-fir	156846
318	sugar maple	118325
611	sweetgum	101484
746	quaking aspen	96715
12	balsam fir	86064
802	white oak	84755
122	ponderosa pine	75202
108	lodgepole pine	67975
111	slash pine	66155
621	yellow-poplar	63965
241	northern white-cedar	62359
129	eastern white pine	53025
833	northern red oak	53022
531	American beech	47944
261	eastern hemlock	47848
832	chestnut oak	43855
827	water oak	42530
125	red pine	41339
762	black cherry	40180

Species	Common Name	n
68	eastern redcedar	40141
97	red spruce	37185
110	shortleaf pine	36776
835	post oak	36269
19	subalpine fir	35695
95	black spruce	35102
693	blackgum	33933
263	western hemlock	33543
375	paper birch	33129
837	black oak	30592
371	yellow birch	30420
15	white fir	29879
544	green ash	29162
93	Engelmann spruce	28152
541	white ash	27270
694	swamp tupelo	26142
543	black ash	25728
17	grand fir	24411
972	American elm	24114
65	Utah juniper	24010
951	American basswood	23279
409	mockernut hickory	23221
403	pignut hickory	22889
121	longleaf pine	21857
971	winged elm	21481
711	sourwood	20378
132	Virginia pine	19950
372	sweet birch	19939
71	tamarack (native)	19866
743	bigtooth aspen	19450
106	common or two-needle pinyon	19395
812	southern red oak	18931
806	scarlet oak	18338
11	Pacific sliver fir	18017
820	laurel oak	16439
407	shagbark hickory	15474
94	white spruce	15266
701	eastern hophornbeam	13759
242	western redcedar	13457
105	jack pine	13414
222	pondcypress	13037

Species	Common Name	n
133	singleleaf pinyon	13011
823	bur oak	12421
653	sweetbay magnolia	11266
264	mountain hemlock	11214
631	tanoak	10615
591	American holly	10425
602	black walnut	9863
391	American hornbeam, musclewood	9556
221	baldcypress	9363
408	black hickory	9106
814	Gambel oak	9020
81	incense-cedar	8940
931	sassafras	8811
402	bitternut hickory	8803
73	western larch	8678
462	hackberry	8646
313	boxelder maple	8179
831	willow oak	8086
805	canyon live oak	7649
461	sugarberry	7557
691	water tupelp	7490
126	pitch pine	7424
838	live oak	7377
491	flowering dogwood	7325
756	honey mesquite	7324
975	slippery elm	7170
741	balsam poplar	6708
61	Ashe juniper	6681
351	red alder	6462
211	redwood	6378
809	northern pin oak	6130
116	Jeffrey pine	5953
20	California red fir	5857
66	Rocky Mountain juniper	5774
901	black locust	5730
912	cabbage palmetto	5728
317	silver maple	5648
64	western juniper	5638
731	American sycamore	5338
69	oneseed juniper	5331
813	cherrybark oak	5272

Species	Common Name	n
107	sand pine	5103

Height Growth Prediction

We fit the equation:

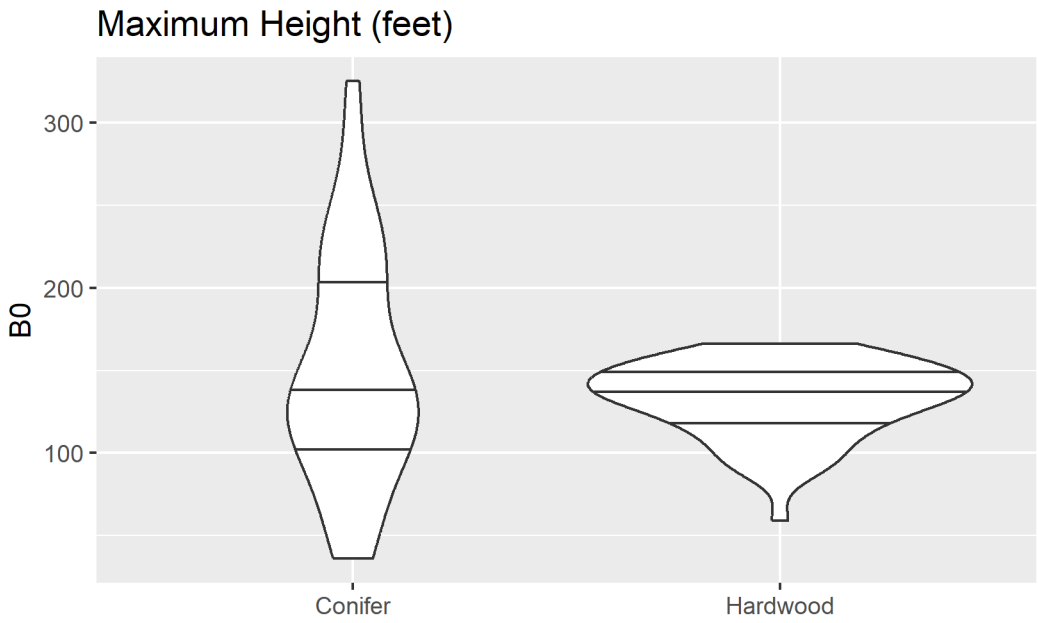
$$\Delta ht = ht_{max}\beta_1\beta_2 cr^{\beta_3} e^{(-\beta_1 ht - \beta_4 ccfl - \beta_5 cch^{0.5} - \beta_6 pc_1 + \beta_7 * pc_2)}(1 - e^{-\beta_1 ht})^{(\beta_2 - 1)} \quad (1)$$

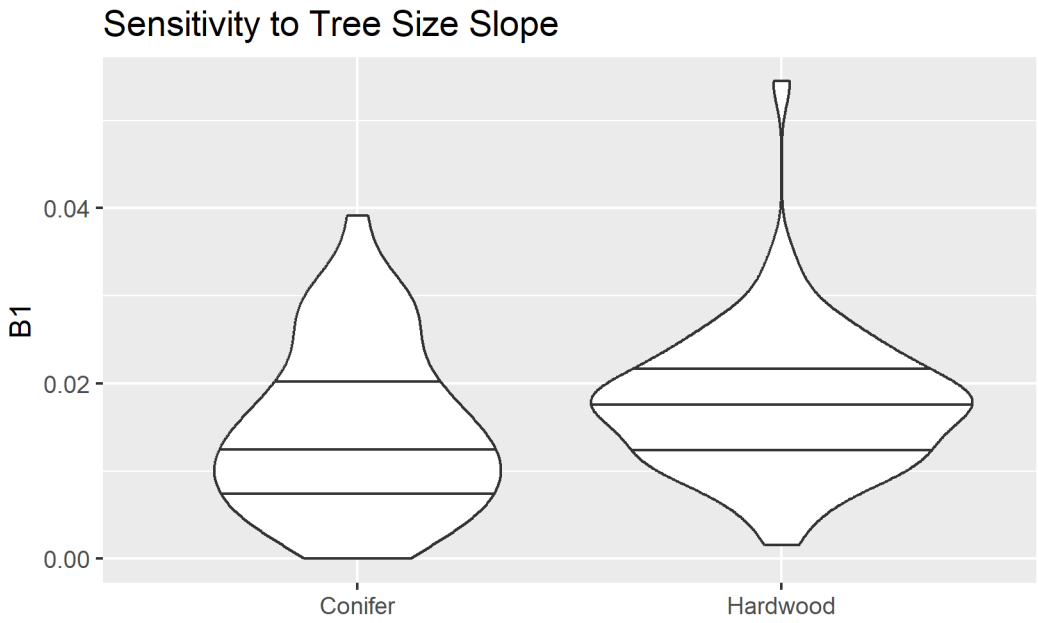
to the all 3495112 growth observations, estimating β_0 through β_7. Where:

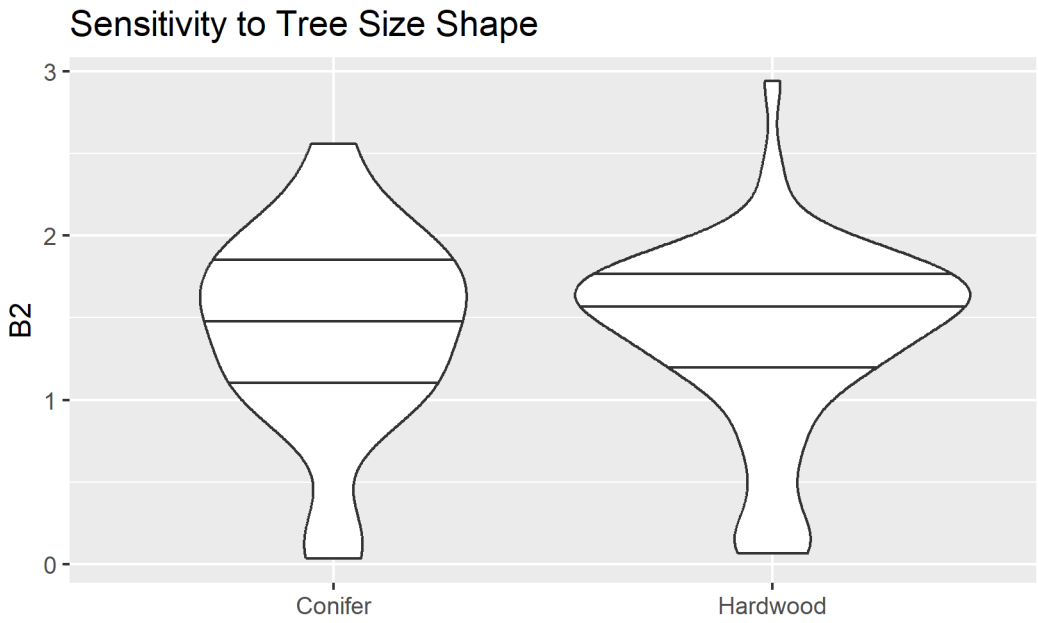
- ht = total height (feet)
- cr = crown ratio (fraction of total height)
- $ccfl$ = crown competition factor in larger trees ($feet^2/acre$)
- cch = crown closure at tree tip (fraction of acre)
- $elev$ = elevation above sea level (feet)
- TD = temperature difference between minimum and maximum ($^\circ C$)
- EMT = extreme minimum temperature ($^\circ C$)
- $pc_1 = 0.252DD_0 + 0.002TD - 0.035PPT_{sm} + 0.967DD_{18}$
- $pc_2 = 0.882DD_0 + 0.015TD + 0.42PPT_{sm} - 0.215DD_{18}$
- DD_0 = growing degree days above $0^\circ C$
- DD_{18} = growing degree days above $18^\circ C$
- TD = Temperature Difference (difference between the mean temperature of the warmest month (MWMT) and the mean temperature of the coldest month (MCMT)
- PPT_{sm} = summer precipitation (mm)

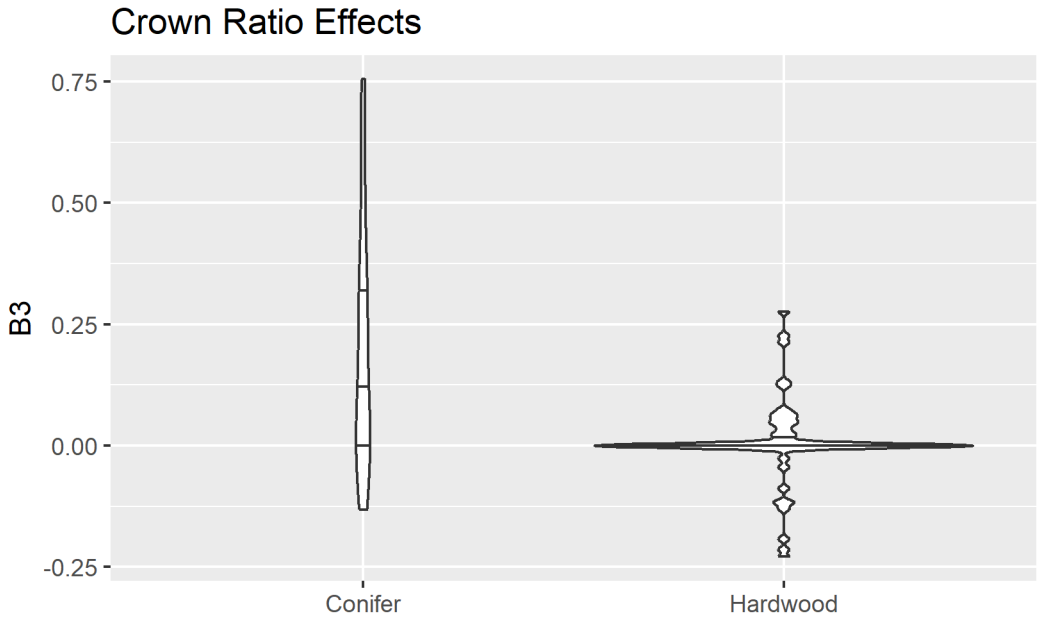
ht_{max} was estimated by finding the maximum height observed in the data set for each species.

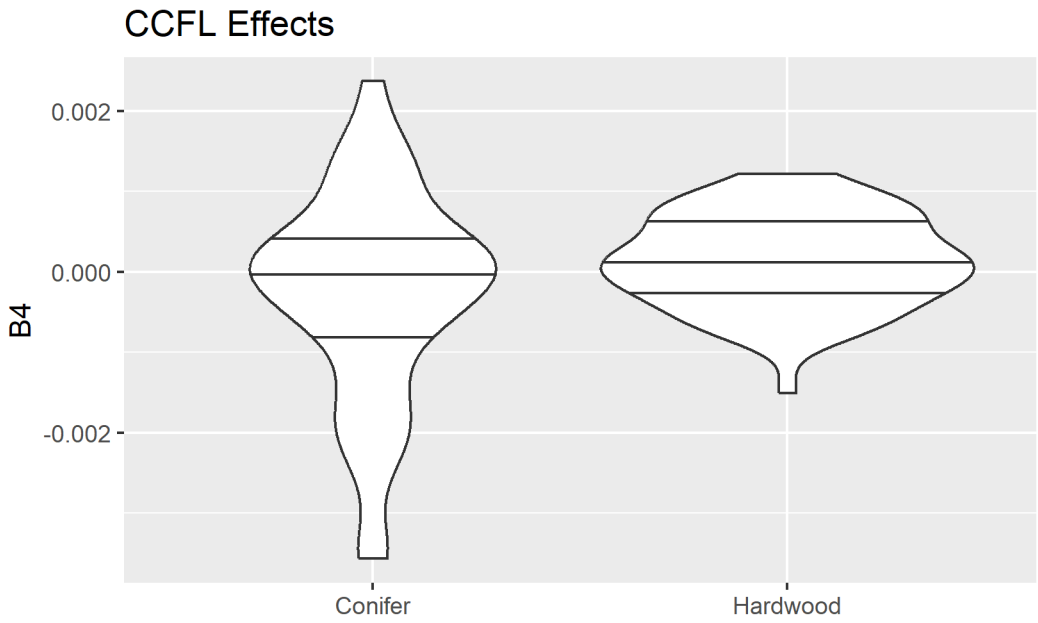
Preliminary CONUS-wide Height Growth Equation

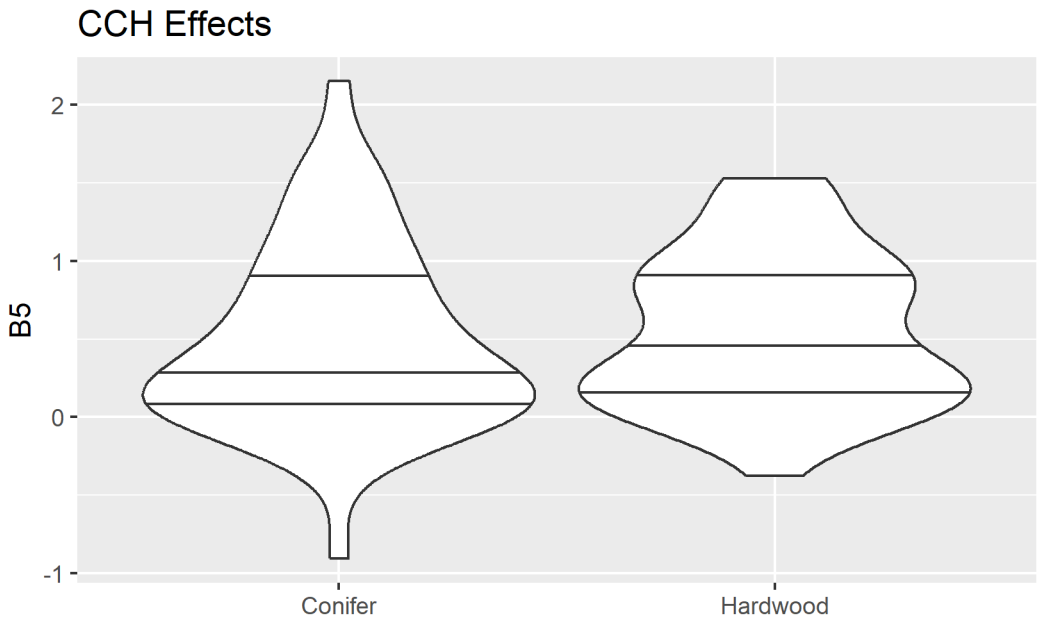


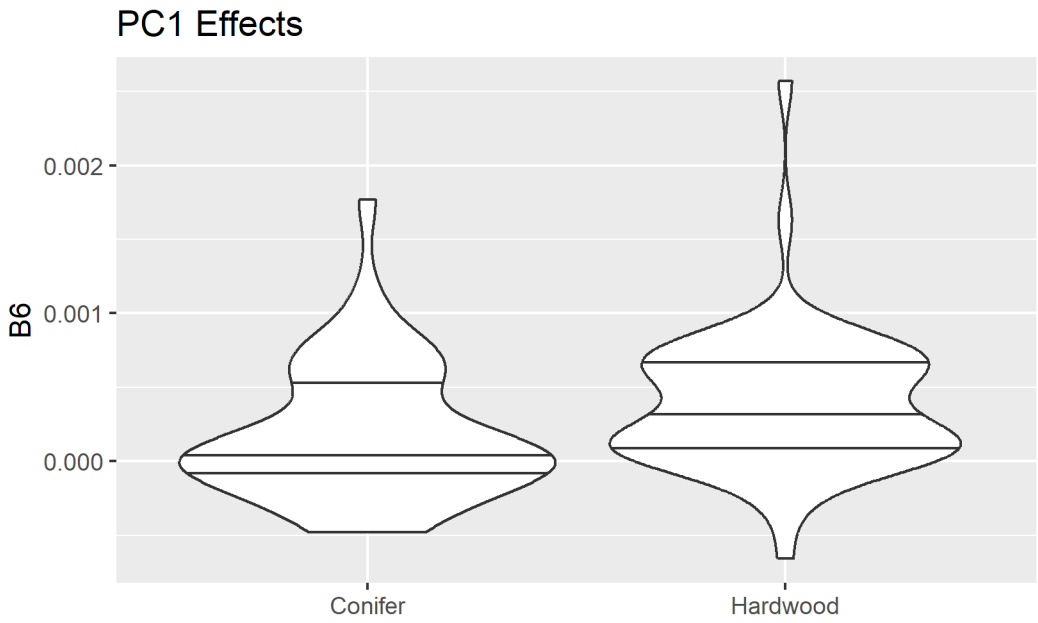


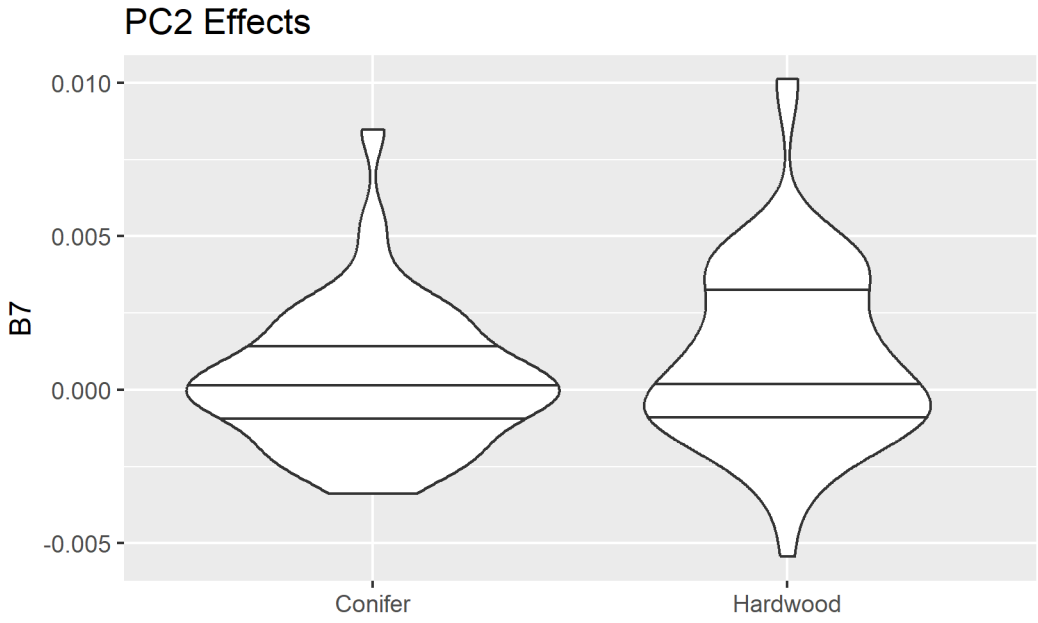

Sensitivity to Tree Size Shape
B2
3
2
1
0
Conifer
Hardwood

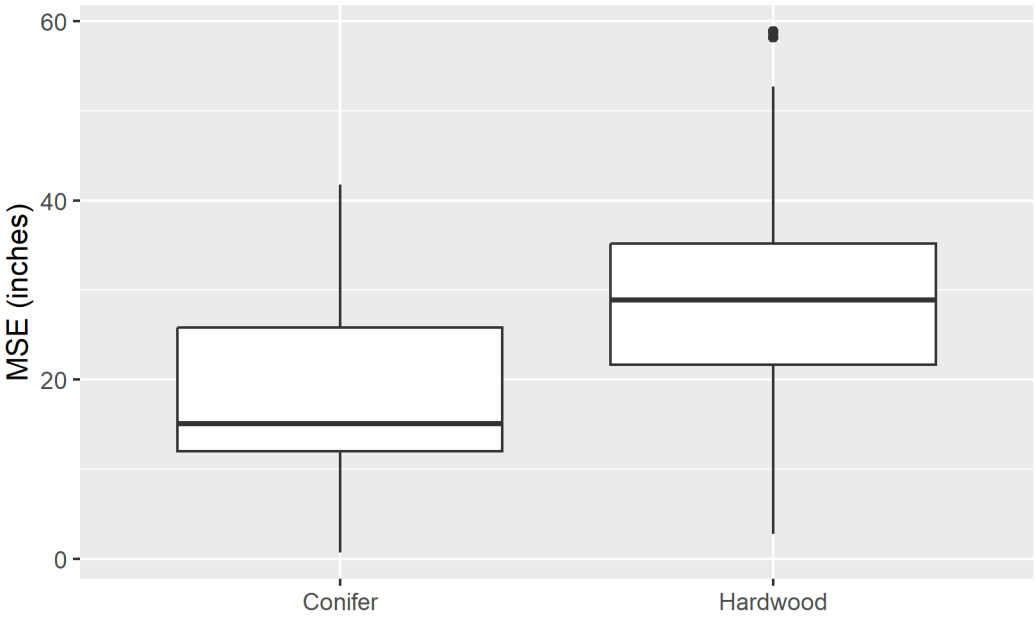

Crown Ratio Effects
B3
0.75
0.50
0.25
0.00
-0.25
Conifer
Hardwood


CCFL Effects
B4
0.002
0.000
-0.002
Conifer
Hardwood

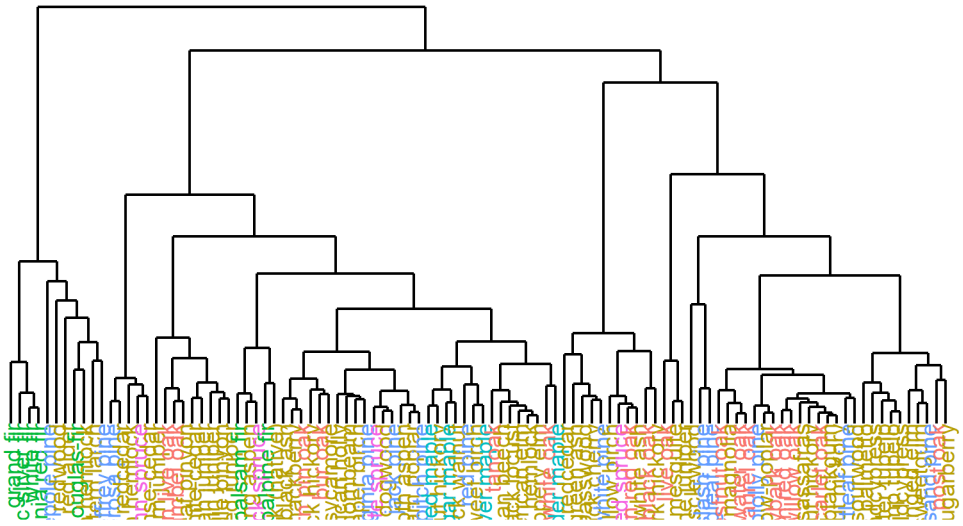

CCH Effects
B5
2
1
0
-1
Conifer
Hardwood


PC1 Effects
B6
0.002
0.001
0.000
Conifer
Hardwood

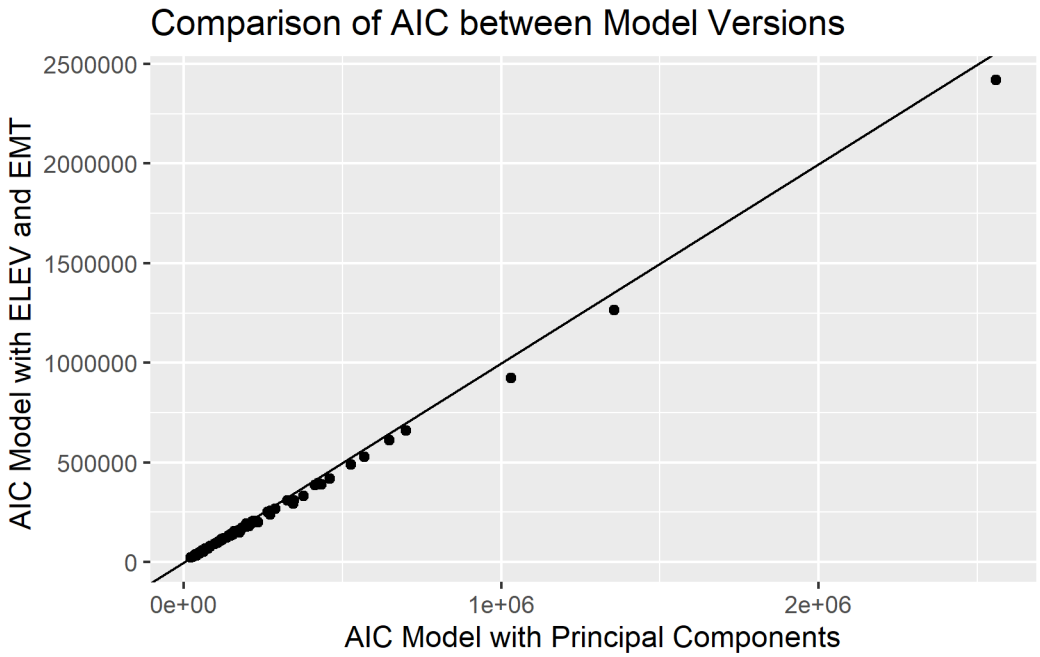

PC2 Effects
B7
0.010
0.005
0.000
-0.005
Conifer
Hardwood



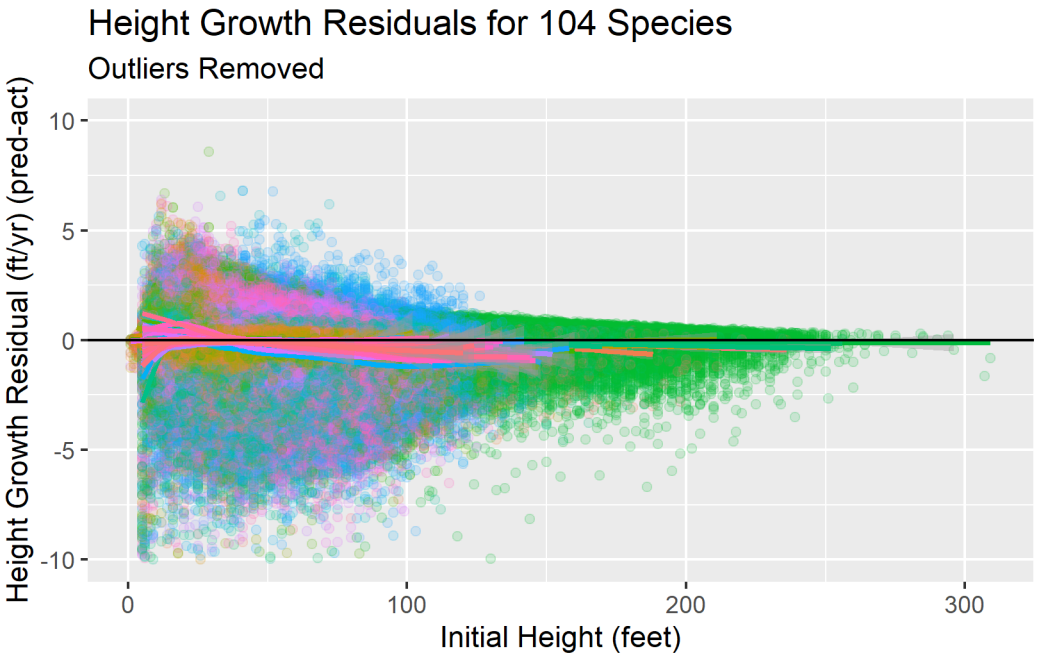
FIA Species Clusters by Height Growth Parameter Estimates

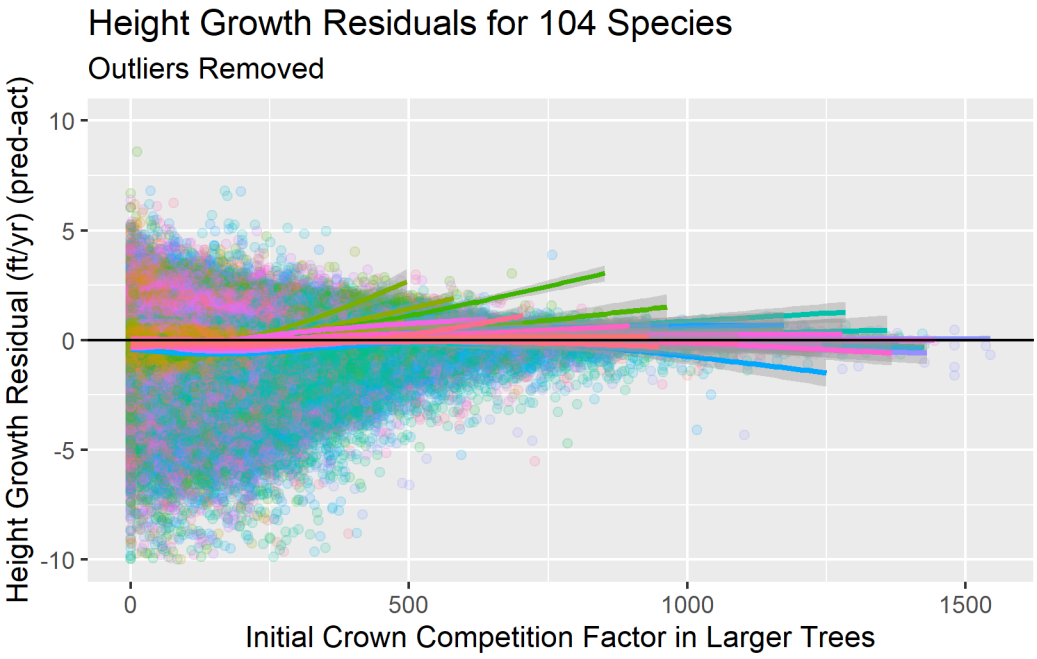


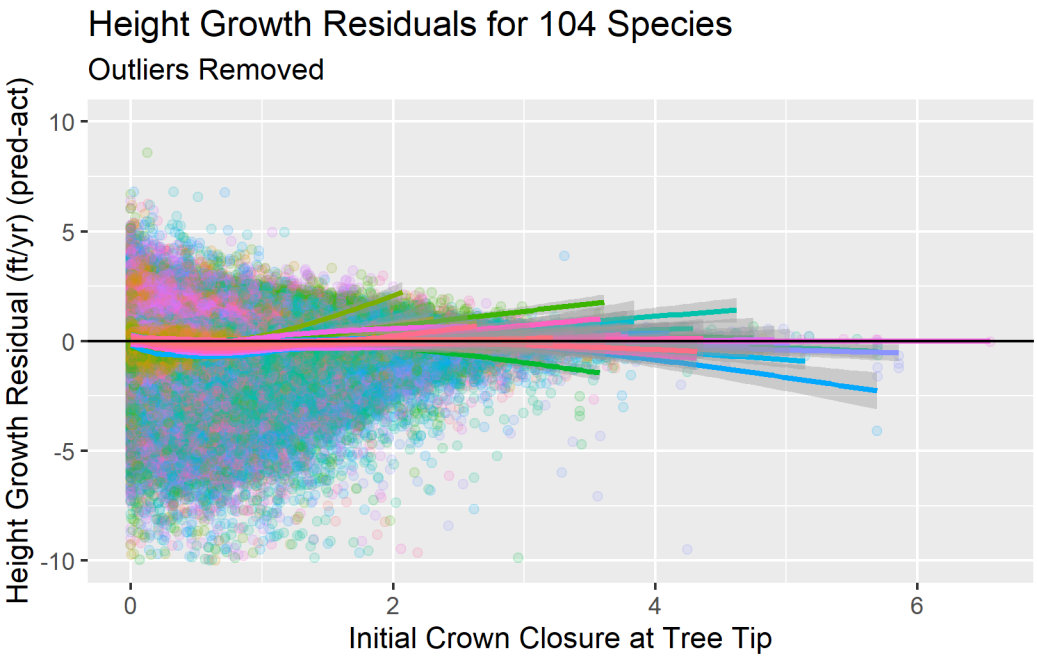
Comparison with Previous Model using Elevation and Extreme Minimum Temperature

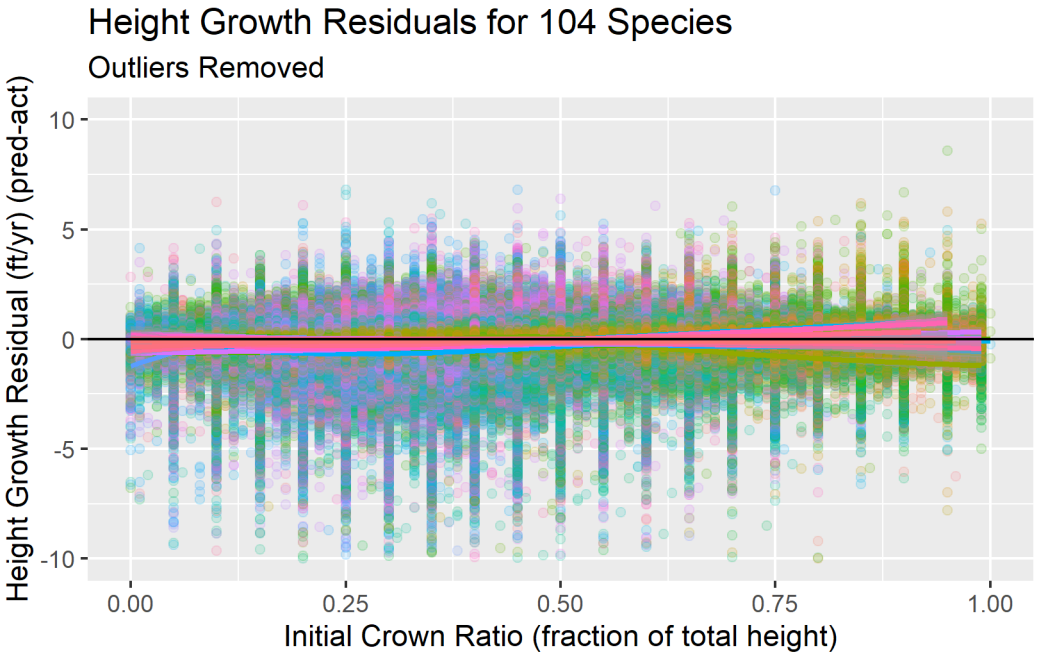


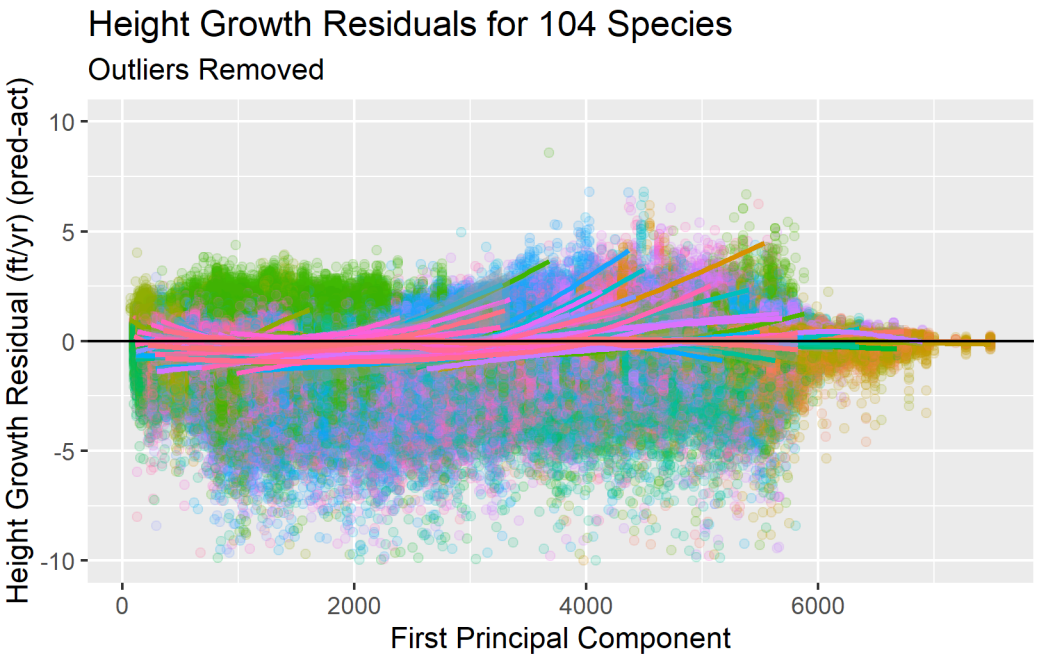
Residual Analysis

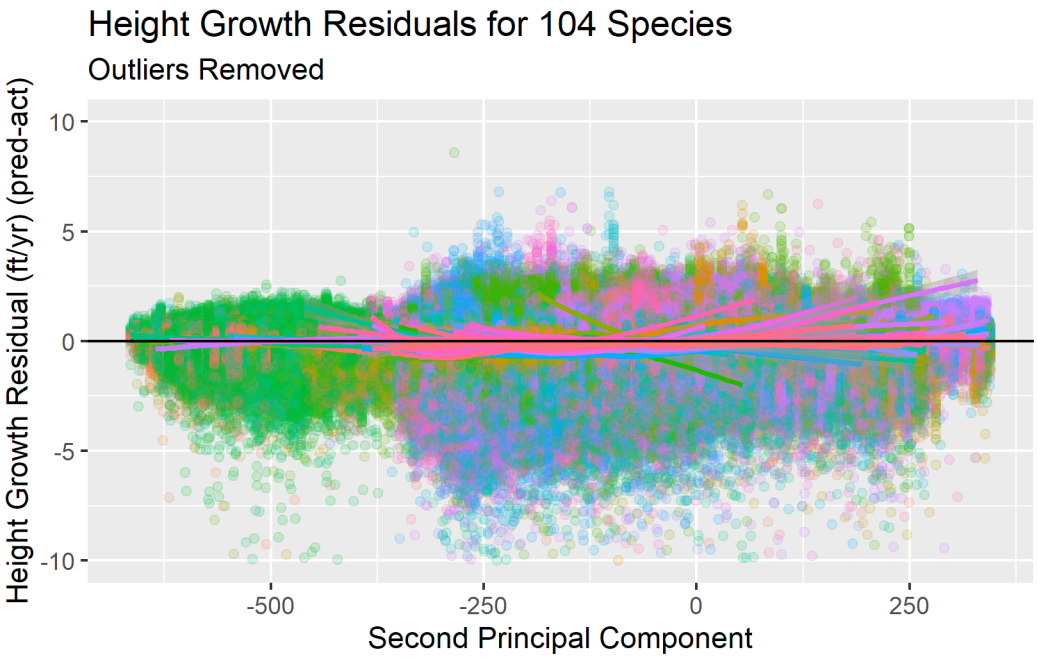


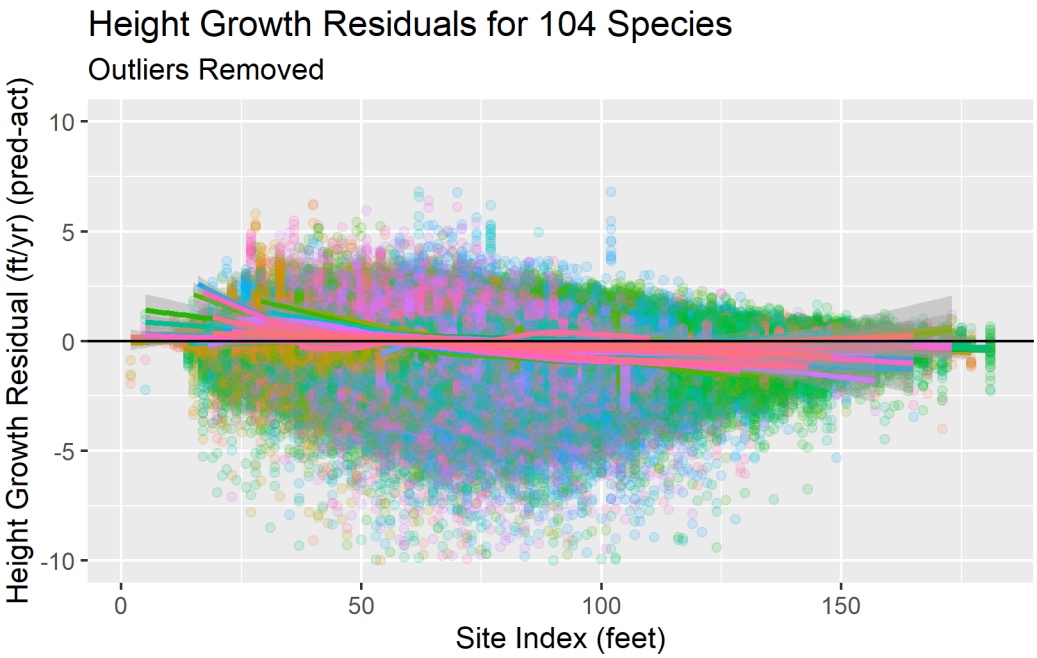



Height Growth Residuals for 104 Species
Outliers Removed
Height Growth Residual (ft/yr) (pred-act)
10
5
0
-5
-10
0
2
4
6
Initial Crown Closure at Tree Tip


Height Growth Residuals for 104 Species
Outliers Removed
Height Growth Residual (ft/yr) (pred-act)
10
5
0
-5
-10
0.00
0.25
0.50
0.75
1.00
Initial Crown Ratio (fraction of total height)


Height Growth Residuals for 104 Species
Outliers Removed
Height Growth Residual (ft/yr) (pred-act)
10
5
0
-5
-10
0
2000
4000
6000
First Principal Component


Height Growth Residuals for 104 Species
Outliers Removed
Height Growth Residual (ft/yr) (pred-act)
10
5
0
-5
-10
-500
-250
0
250
Second Principal Component


Height Growth Residuals for 104 Species
Outliers Removed
Height Growth Residual (ft/yr) (pred-act)
10
5
0
-5
-10
0
50
100
150
Site Index (feet)